

Tianhao Hu

RESEARCH INTEREST

**Numerical PDE,
Machine Learning,
Inverse Problems,
Scientific Computing**

CONTACT

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PERSONAL WEBSITE

Personal website:

<https://hhjc-web.github.io/>

Google Scholar:

<https://scholar.google.com/citations?user=D7NWc1QAAAAJ&>

GitHub: <https://github.com/hhjc-web>

PUBLICATION LIST

- T. Hu, B. Jin, and Z. Zhou, Solving elliptic problems with singular sources using singularity splitting deep Ritz method, SIAM Journal on Scientific Computing, 45 (2023), pp. A2043-A2074.
- T. Hu, B. Jin, and Z. Zhou, Solving Poisson problems in polygonal domains with singularity enriched physics informed neural networks, SIAM Journal on Scientific Computing, 46 (2024), pp. C369-C398.
- T. Hu, B. Jin, and F. Wang, An iterative deep Ritz method for monotone elliptic problems, Journal of Computational Physics, 527 (2025), pp.113791.
- T. Hu, B. Jin, and Z. Zhou, Point source identification using singularity enriched neural networks, IMA Journal of Numerical Analysis, page draf129, 02 2026.
- T. Hu, X. Huang, B. Jin, Q. Quan, and Z. Zhou, Conditional Stability and Numerical Reconstruction of a Parabolic Inverse Source Problem Using Carleman Estimates, Inverse Problems, minor revision, 2026.
- T. Hu, B. Jin, and Y. Wang, Stability of Electrical Impedance Tomography with Anisotropies and its Application to the Deep Caldeón Method, SIAM Journal on Imaging Sciences, in press, 2026.
- T. Hu, G. Li, F. Wang, Y. Xu, and Z. Zhou. Dual variational neural network for the p-Laplace problem. arxiv.org: 2605.28291, submitted to Journal of Computational Physics, 2026.
- T. Hu, B. Jin, and Y. Wang. A Convolutional Deep Calderón Method for Electrical Impedance Tomography, submitted to SIAM Journal on Scientific Computing, 2026.
- T. Hu, B. Jin, F. Wang and Y. Xu. A Hybrid Iterative Deep Ritz Method for Elliptic Interface Problems, submitted to Journal of Computational Physics, 2026.

ONGOING PROJECTS

- T. Hu, B. Jin, and F. Wang, A Level-Set Random Feature Petrov–Galerkin Method for Elliptic and Parabolic Interface Problems.
- T. Hu and Z. Zhou, A recurrent Fourier Neural Network to Recovery Two Function-valued Coefficients in the Helmholtz Equation for Inverse Scattering Problems.
- T. Hu, B. Jin, Y. Wang, Learning Solution maps from Linear Wave equations to Nonlinear wave equations.
- J. Han and T. Hu, Diffusion-Based Out-of-Distribution Detection for Active Learning in Scientific Machine Learning.

EDUCATION

- 09/2019-06/2023, B.Sc., School of Mathematics, Jilin University.
- 08/2023-07/2027(estimated), Ph.D., Department of Mathematics, The Chinese University of Hong Kong.

CONFERENCES

- Invited Speaker, talk "Solving Elliptic Problems with Singular Solutions Using Splitting Technique", at The Second Hong Kong SIAM Biennial conference, Hong Kong, 28th Aug – 1st Sep, 2023.
- Invited Speaker, talk "Identifying Point Sources in Elliptic Problems using Deep Neural Networks", at SIAM Conference on Computational Science and Engineering, Fort Worth, Texas, U.S., 3rd-7th Mar, 2025.
- Invited Speaker, talk "Point Source Identification using Singularity Enriched Neural Networks", at The 14th Annual Conference on Inverse Problems, Imaging, and Applications, Tianjin, China, 18th-21st Apr, 2025.
- Invited Speaker, talk "Dual variational neural network for the p-Laplace problem", at The 15th AIMS Conference, Athens, Greece, 6th-10th Jul, 2026.

GRANTS

- 2026, National Natural Science Foundation of China, Project 125B2022, ¥ 300,000, PI.

RESEARCH VISITS

- 10/2026-12/2026, Center for Computational Mathematics, Flatiron Institute, New York, Host: Prof. Jiequn Han.

AWARDS

SCHOLARSHIPS

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| (i) 2020,2021,2022 | National scholarship | Ministry of Education, China |
| (ii) 2022 | Samsung Scholarship | Samsung (China) Investment Corporation |
| (iii)2022 | Hongqi Scholarship | China FAW Group Corporation |
| (iv)2023 | CUHK Vice-Chancellor's Ph.D. Scholarship | The Chinese University of Hong Kong |
| (v) 2024 | Madam Wong Siew Khoon Scholarship | The Chinese University of Hong Kong |

RECOGNITIONS

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| (i) 2026 | CAST Youth Talent Cultivation Project – PhD Special Program | China Association for Science and Technology |
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TEACHING EXPERIENCE

- Teaching Assistant, MATH 1510G Calculus for Engineers, 2023 Fall.
- Teaching Assistant, MATH 3270B Ordinary Differential Equations, 2024 Summer.
- Teaching Assistant, MATH 3230A Numerical Analysis, 2024 Fall.
- Teaching Assistant, MATH 3340 Mathematics of Machine Learning, 2025 Spring.
- Teaching Assistant, MAAT 5391 Mathematical Theories of Machine Learning, 2025 Fall.
- Teaching Assistant, MATH 3340 Mathematics of Machine Learning, 2026 Spring.
- Teaching Assistant, MAAT 5392 Mathematical Principles of Artificial Intelligence, 2027 Spring.
- Teaching Assistant, MATH 3340 Mathematics of Machine Learning, 2027 Spring.